UPPSALA
UNIVERSITET

Institutionen för samhällsbyggnad och Industriell teknik
UPPSALA UNIVERSITY

Industrial Project Management (1TE061) EXAM

17th March 2020.

This exam consists of 8 questions and you can earn a maximum of 40 points. To pass the exam, you will need at least 18 points. The first set of questions (Q1-2) will only require very short answers. The second set of questions (Q3-6) will require you to explain something in-depth or perform simple calculations. The third set of questions (7-8) are essay-type questions and will require reasoning and reflection from what you have learnt during the course.

Please note: Use a new sheet for every new question and write your exam code on each new sheet.
Permitted aids: Simple calculators and English dictionary (not digital) are allowed – no other notes or books can be brought to the exam.

Max: 40p
3: 18p (45%)
4: 28p (70%)
5: 34p (85%)

Good luck

Simon & Per

Provided formulas

$$Cost\ Performance\ Indicator\ (CPI) = \frac{Earned\ Value}{Actual\ Cost}$$

$$Schedule\ Performance\ Indicator\ (SPI) = \frac{Earned\ Value}{Planned\ Value}$$

1. **Pre – study – Part A**
    a. What is a **goal seeking project** and **goal-oriented project**? (1 point)
    b. What is the **100% rule** of the **Work Breakdown Structure** (**WBS**)? (1 points)
    c. Define what is a **project goal** and **Impact goal**? (1 point)

2. **Pre- study – Part B**
    a. What is a **project charter**? (1p)
    b. What is **SWOT analysis** and how is it applied during the pre-study phase? (2 point)
    c. What is the main distinction between a **Strong Matrix** and **Weak Matrix**? (1 points)

3. **Project planning, roles and tools**
    a. Please explain the concept of **Scope management**. (1 point)
    b. List at least 2 roles of a **steering committee** during the project. (1 points)
    c. Define the meaning of the following tools and briefly state how they are applied during the planning stage of a project: **EVM- Earned Value Management**, **PERT- Program Evaluation and Review Technique**, **CPM – The Critical Path Method.** (6 points)

4. **Earned Value Management (EVM).**
    a. Explain briefly what is **planned value** and **Earned Value**. (1 point)
    b. The table below shows planned cost (**Budget**) and planned time consumption (**Duration**). Assume that the project is now in **month 5**, and only (1- 4) activities are completed and you have so far spent € 36000. With the information given in the table below compute and calculate the following.
        I. What is the **CPI- Cost Performance Indicator**? (1 point)
        II. What is the **SPI- Schedule Performance Indicator**? (1 point)
        III. What can you conclude from the **CPI** and **SPI** you have found above? (1point)

Activity	**Duration**	**Budget**
1	1 month	€ 3000
2	1 month	€ 6000
3	1 month	€ 7000
4	1 month	€ 11000
5	1 month	€ 14000
6	1 month	€ 16000
Total:	6 months	€ 57000

5. **Critical Path**

Below is a list of activities of a work package that belongs to a Digital supply chain project. With the information below and duration in weeks. You are to construct a logical network, perform a forward and backward pass to show the critical path. Show clearly how to compute for total float / slack. Please show your working all through and indicate the final critical path in bold. (5 Points)

Activities	Description of task	Duration	Preceding Activity
A	Project group composition	10	none
B	Operational data requirements	5	A
C	Innovative tasks to utilize Blockchain	25	A
D	Digital Exchange platform with AI	20	B
E	Digital integration with user cases	50	C
F	Data security protocols tested	15	C
G	Final test with simulations	35	D
H	Validation and user acceptance	15	E, F, G

6. **Project closure & hand-over.**
    a. During project closure auditing and review is important, briefly explain what is considered in **auditing** and **review**. (1 points)
    b. Reflect on the **closure phase** of a project. Why is it important and how can you facilitate it? (3 points)

7. **Creating project group and leadership skills in projects**

    a. SCANIA AB is a major Swedish manufacturer of commercial vehicles such as heavy trucks and busses. Due to their sustainability agenda, SCANIA will start a new project focusing on alternative fuels and electrification engines. The project has attracted several competences and from different cultures. Almost 80 % of the people in the project have never worked together. Martha who is project manager is wondering how ensure functional collaboration, setting up a framework for communication and how all these personalities can be managed to ensure that the project finishes on time. Please reflect on behavior style models such Belbin's model, DISC, FIRO, or leadership & communication skills and document some principles that Martha could apply to ensure that a project finishes successfully. (6 Points)

8. **Risk, uncertainty & Quality**
    a. All projects encompass a certain measure of risk and uncertainty. Given that you are project manager of large IT project. Please document in not more than one page, what to include in your risk management plan and discuss briefly how risks may affects quality? (6 points)